

DJI Agras – Troubleshooting Agras Drones 2022

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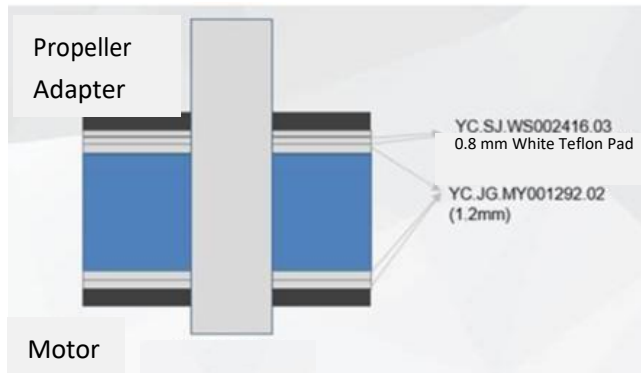
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1. Introduction to T40/T20P Propeller Gasket

Phenomenon: There are two types of T40/T20P propeller gaskets.

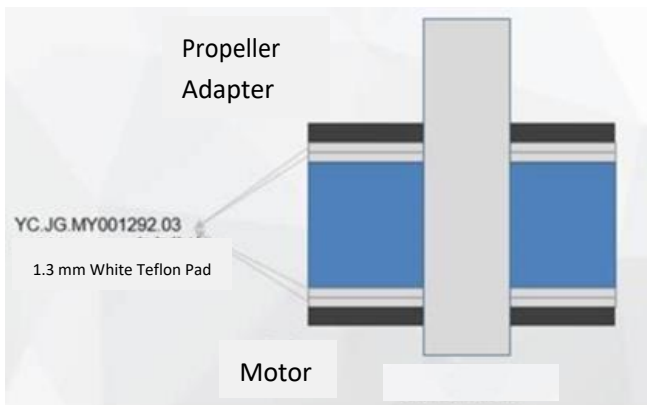
Type 1:

Gaskets include: 1.2 mm Teflon Pad (YC.JG.MQ001292.02) ×3 + 0.8 mm Teflon Pad (YC.SJ.WS002416.03) ×2 (0.8 mm Teflon Pad ×2 + 1.2 mm White Teflon Pad ×1 are on the upper part; 1.2 mm White Teflon Pad ×2 are on the lower part)



● Type 2:

Gaskets include: 1.3 mm Teflon Pad (C.JG.MQ001292.03) ×4 (two pads on the upper part and two on the lower part)



- Solution: When replacing the propellers, if there is a 1.3 mm Teflon Pad (C.JG.MQ001292.03) on the original product, please use the propeller gaskets following the Type 2.

2. What should I do if the app prompts a 103/105422 error code and the T series Agras drone cannot be activated?

- **Situation 1:** The issue occurs when activating the replacement drone.
- **Troubleshooting Plan:** Provide the repair case number and product's serial number, and contact DJI after-sales team.
- **Situation 2:** The issue occurs when activating the drone for the first time after **its aerial-electronics module is replaced during the repair**.
- **Cause:** The serial number of the aerial-electronics module shown in the DJI Agriculture Management Platform is not the same with that of the aerial-electronics module for the drone.
- **Solution:**
Check whether the new serial number that is used to **replace the original serial number of the aerial-electronics module in the Dealer system** is consistent with the serial number of the aircraft aerial-electronics module, and **make sure that the new serial number is input correctly**.
Provide the screenshot of the aerial-electronics module **serial number** on the **DJI Agras app** and the **case number** in the Dealer system, and contact the DJI agricultural after-sales service manager.

3. What T20P components are compatible with T30 drone?

Intelligent Flight Battery	T20P	T30
T20P Intelligent Flight Battery	Yes	Not applicable
T30 Intelligent Battery	Yes, and flight time increased	Yes
Generator/Charger	T20P Intelligent Flight Battery	Intelligent Flight Battery
D12000iE Multifunctional Inverter Generator	Yes (adaptive power)	Yes (adaptive power)
D6000i Multifunctional Inverter Generator	Yes	Yes, but slow charging
T20P Intelligent Charger	Yes	Yes, but slow charging
D9000i Multifunctional Inverter Generator	Yes (firmware update required)	Yes
T30 Intelligent Charger	Yes (firmware update required)	Yes
Smart Controller	T20P	T30
T20P Smart Controller	Yes	Not applicable
Propellers	T20P	T30
T20P Propellers	Yes	Not applicable

4. T40/T20P Delivery Pump Worn, Scorched, Shaft Broken Issues

- **Phenomenon:** The T40/T20P delivery pump is worn, scorched or the shaft is broken, and the issues mainly occur on the samples.
- **Solution:** Replace the **impeller pump module** (BC.AG.SS000644.01). The module includes the impeller module and pump shell module.

5. The T40/T20P motor cannot rotate smoothly.

- **Phenomenon:** The T20P motor cannot rotate smoothly.
- **Solution:** Replace the motor and check **whether there are debris inside the motor**. If yes, check there are debris inside the other motors, and replace them in time.
- **Background:** Water flushing will let the water in the specific position of the T40/T20P motor, which will cause the bearing to rust after using for a long time and the motor cable will be stuck.



Figure 1 Avoid flushing water into the position as shown.



Figure 2 The water may enter into this part as shown.

- **Phenomenon:** After flushing the aircraft, leave it for a period of time, and rotate the motor by hand. It is found that the motor cannot rotate smoothly (the bearing is water damaged and rusty).
- **Note:** Advise users not to directly flush to the position shown in the figure 1 (you can wipe it off if there are residual liquid);
- **The following information need to be provided:** Please record the information according to the template when the motor is stuck.
 - a) Describe whether it is the motor on the upper position or lower position, and provide the motor's serial number.
 - b) Actual operating time and conditions: For example: 14:50 Feb 21, 2022, fertilization for wheat
 - c) Describe whether the drone was washed.
 - d) Describe how the drone is stored (carrying case or indoor environment).
 - e) Describe whether there are any warnings shown in the Agras app.

6. T20 Aircraft Linking Failure

- **Phenomenon:** The T20P drone cannot be linked with the remote controller.
- **Solution:**
 1. Please make sure that the firmware of the aircraft and remote controller is **updated to the latest version** (v01.00.0108 and above).
 2. Replace the drone or remote controller and try linking again.
 3. After the drone is replaced, if linking is successful, check whether the **connection between the drone's SDR**

antenna, RF module, and aerial-electronics module is normal. Then try to replace the RF module and aerial-electronics module in turn for troubleshooting.

4. After the remote controller is replaced, if linking is successful, **check whether the connection between the remote controller's SDR board and main board is normal**. Then try removing and attaching the flexible flat cable connecting the SDR board and main board again for troubleshooting.
5. If none of the above solutions works, export the QR code data, and contact DJI after-sales team for data analysis.

7. T20P Dual Atomized Sprinkler Leakage

- **Phenomenon:** The T20P dual atomized sprinkler is leakage after spraying is turn off.
- **Cause:** It is easy to accumulate pesticides under the **spinner disk gasket**, and the gasket may be **swollen and worn**; The one-way valve cannot intercept the residual moisture due to pesticides accumulation.
- **Solution:**
 1. The one-way valve is stuck or malfunctioning. In this case, it is recommended to **remove the hose and clean it**, or try to replace the valve.



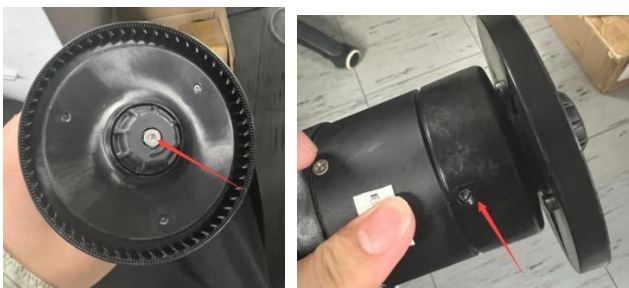
2. The spinner disk gasket is **worn due to residual pesticides under the gasket**. It is recommended to replace the gasket and clean the spinner disk (the gasket is included in the accessory kit).



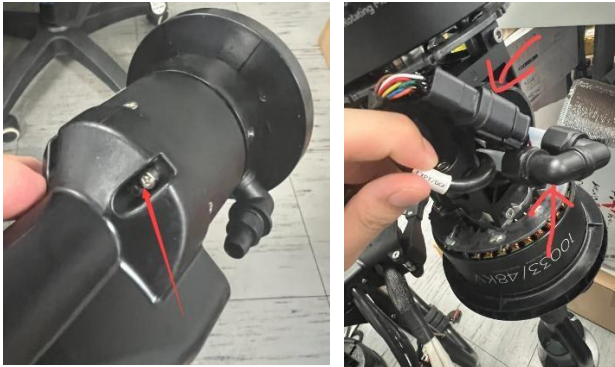
3. If none of the above solutions works, export the QR code data, and contact DJI after-sales team for data analysis.

8. There is a self-check error with the T40/T20P centrifugal ESC/The voltage is too low/The rotational speed is not displayed under Spraying System Settings in the HMS.

- **Disassembly & Assembly Method:**
 1. Remove the hand screw at the bottom of the centrifugal sprinkler, rotate to remove the bottom cap, and then remove the spinner disk, as shown in the left figure.
 2. Remove the screw below the motor and then remove the motor lower cover, as shown in the right figure.



3. Remove the screw securing the motor upper cover, and then remove the upper cover, as shown in the left figure.
4. Remove the motor protective cover, pull out the centrifugal motor cable, remove the centrifugal spray rod sleeve (be sure to fix the spray rod), and pull out the hose curving connector, and then remove the spray rod.



● **Troubleshooting Steps:**

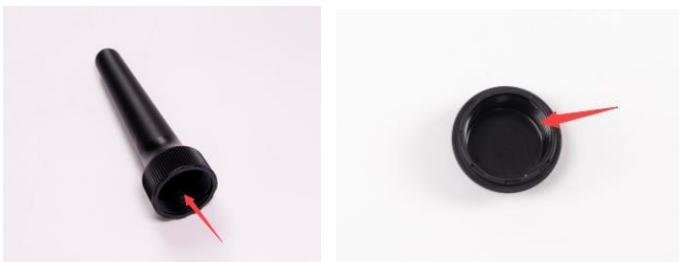
1. Check whether the spinner disk and motor **lower cover are worn or scorched**.
2. **Remove and attach the cable again**. Then check whether the cable connection is normal.
3. If no error is found, replace the dual atomized sprinkler. If the app prompts the sprinkler is malfunctioning, try to install it to the other position. If the warning is still displayed, replace the sprinkler motor. After the sprinkler motor is replaced and the issue persists, it is recommended to **replace the aircraft arm cable and power distribution board**.
4. If none of the above solutions works, export the QR code data, and contact DJI after-sales team for data analysis.

9. T20P SDR Protective Sleeve or Motor Shaft Cover Loose

- Phenomenon: The T20P SDR protective sleeve or motor shaft cover is loose.

● **Solution:**

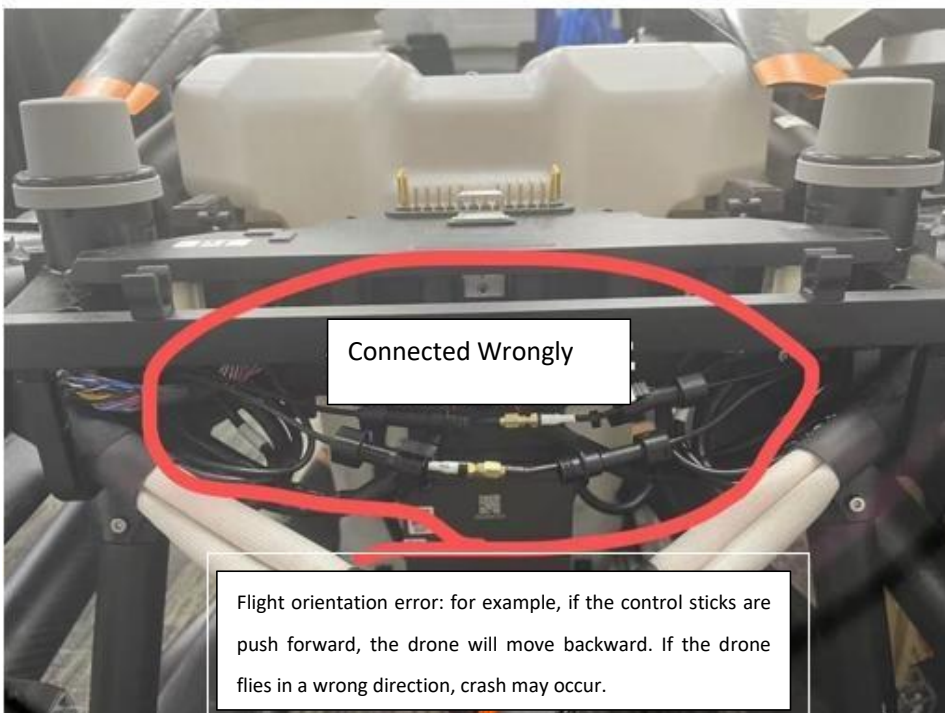
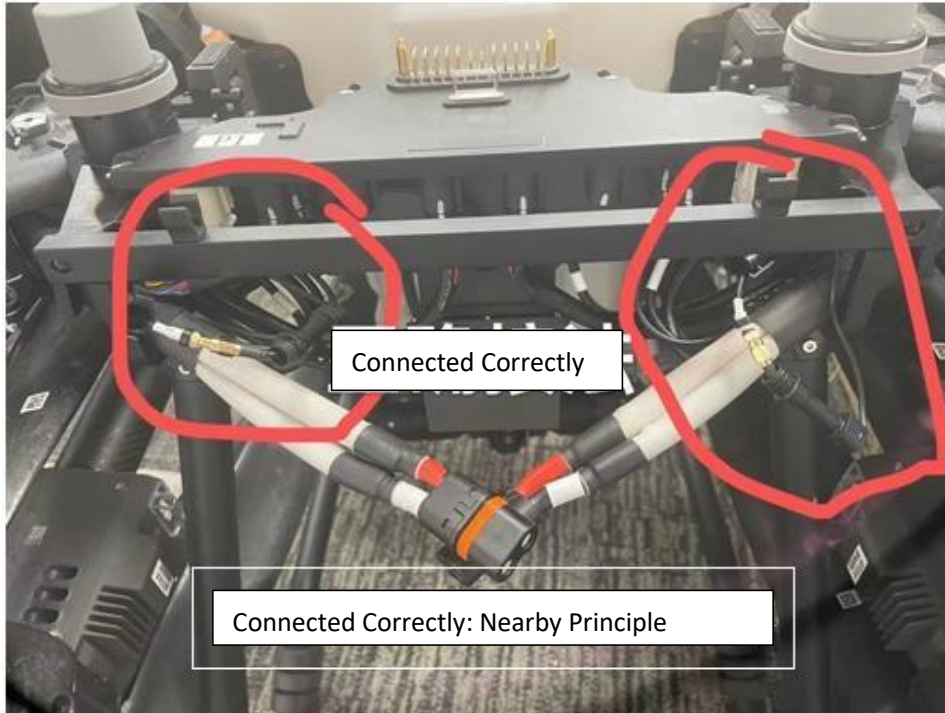
1. Apply a circle of glue to the inner thread of the SDR protective sleeve (YC.JG.ZS002416.03) or motor shaft cover (YC.JG.ZS002292.02).



2. Glue Model: Kafuter K-5911B Glue

10. T20P RTK Antenna Connection Method

T20P RTK Antenna Connection Method: The RTK antennas need to be connected following the nearby principle, that is, the RTK antenna should be connected to the other antenna closer to it. If the RTK antennas are connected wrongly, **there may be an error with the flight orientation**. For example, if the control sticks are push forward, the drone will move backward. If the drone flies in a wrong direction, crash may occur. So please make sure that the RTK antennas are connected correctly.



11. What should I do if the remaining liquid payload in the T20P spray tank is shown 0?

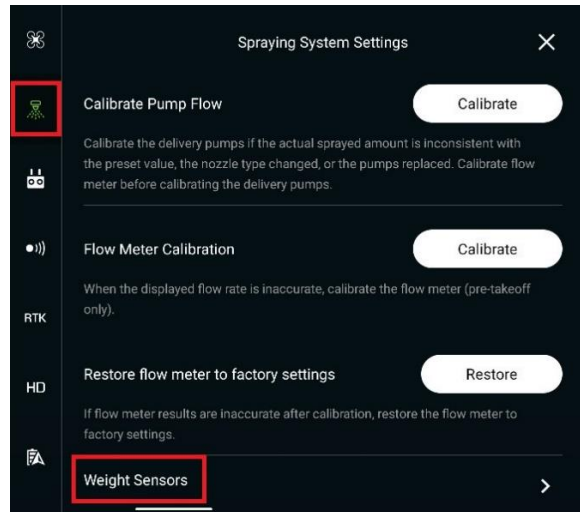
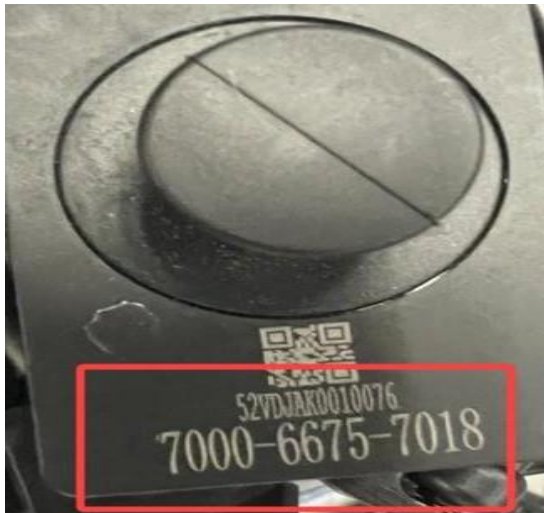
- **Phenomenon:** The remaining liquid payload in the T20P spray tank is shown 0.
- **Troubleshooting Plan:**
 1. If the remaining liquid payload is displayed 0, the Tare Calibration is successful, but the Weight Calibration fails, or the remaining liquid payload is displayed 0 after filling in the water, it can be determined that the **liquid level gauge is damaged**. It is recommended to replace the **liquid level gauge and hall sensor** (at the bottom of the spray tank). The problem is prone to occur after unboxing, and it is suspected that the issue is caused by transportation.
 2. If none of the above solutions works, export the QR code data, and contact DJI after-sales team for data analysis.



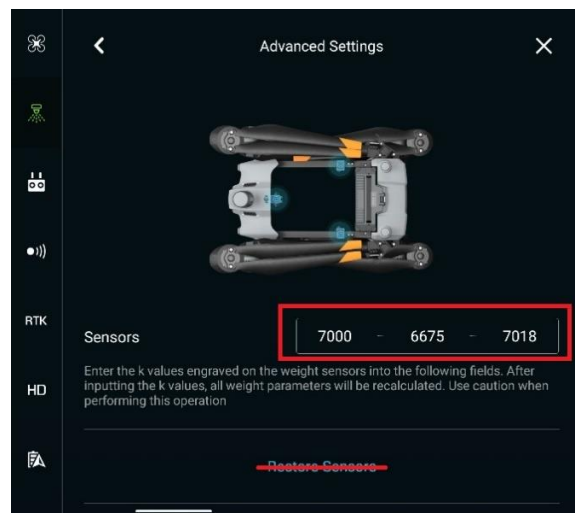
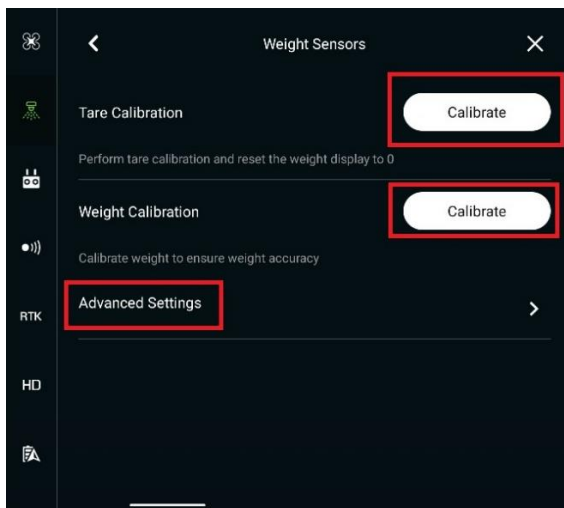
12. What should I do if the value measured by the T40/T20P weight sensor keeps changing?



- **Phenomenon:** The issue must occur when the T20 weight sensor or spraying board is replaced, and there is small probability that the issue occurs after system update. When the weight sensor is reset, and the K value is 1 by default, which causes the issue to occur.
- **Solution:**



Input K value again: Go to DJI Agras app, Spraying System Settings, Weight Sensors, Advanced Settings, then input the K value and tap the confirm button and “x” icon to exit. (**Do not tap “Restore Sensors”.**)



Tare Calibration: Empty the spray tank, and perform Tare Calibration.

Weight Calibration: Add **20 kg** (it needs to be weighed in advance) of liquid into the spray tank, then confirm whether the remaining liquid payload displayed on the remote controller is between **16 kg and 24 kg**. Tap “**Weight Calibration**” in the app to complete the calibration.

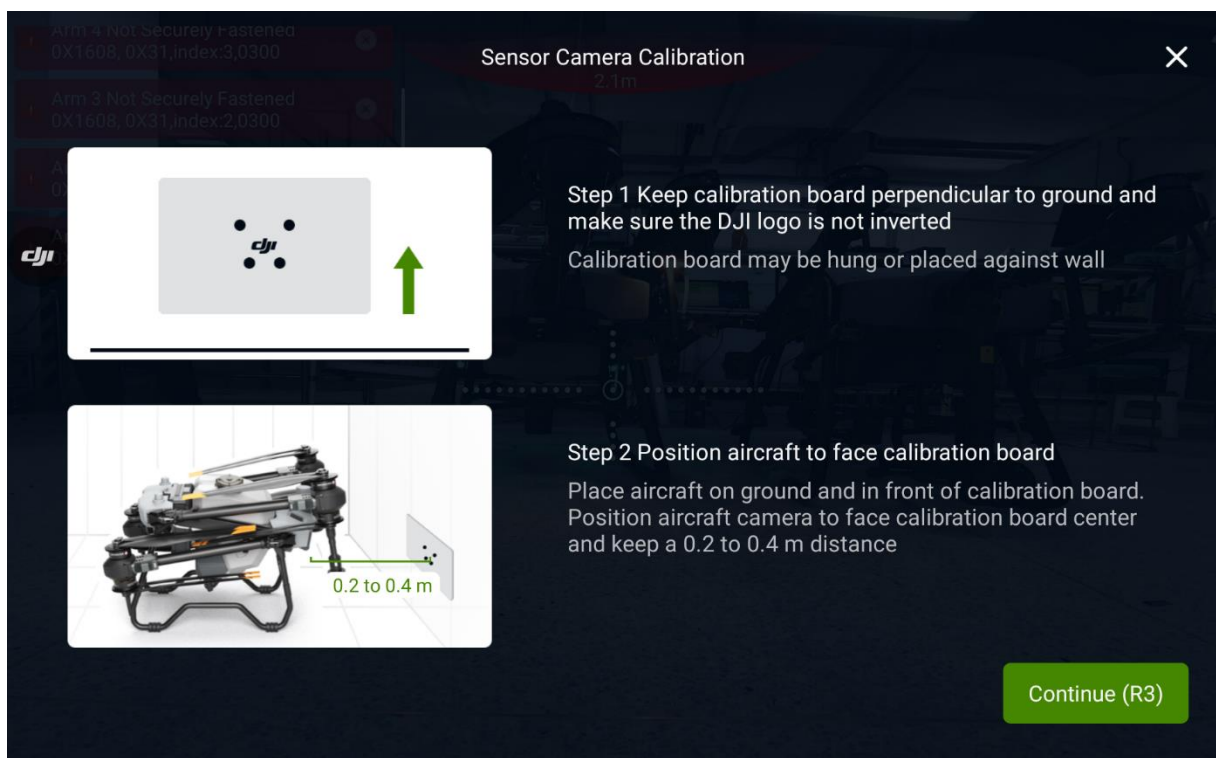
13. Where is the serial number (SN) of the DC module of the T20P?

- **SN Position:** To prevent the SN sticker from falling off if it is attached to the outside, the sticker is located inside the side of the DC module. Open the corresponding cover and then you can find it. The first three digits of the serial number is 5D3.

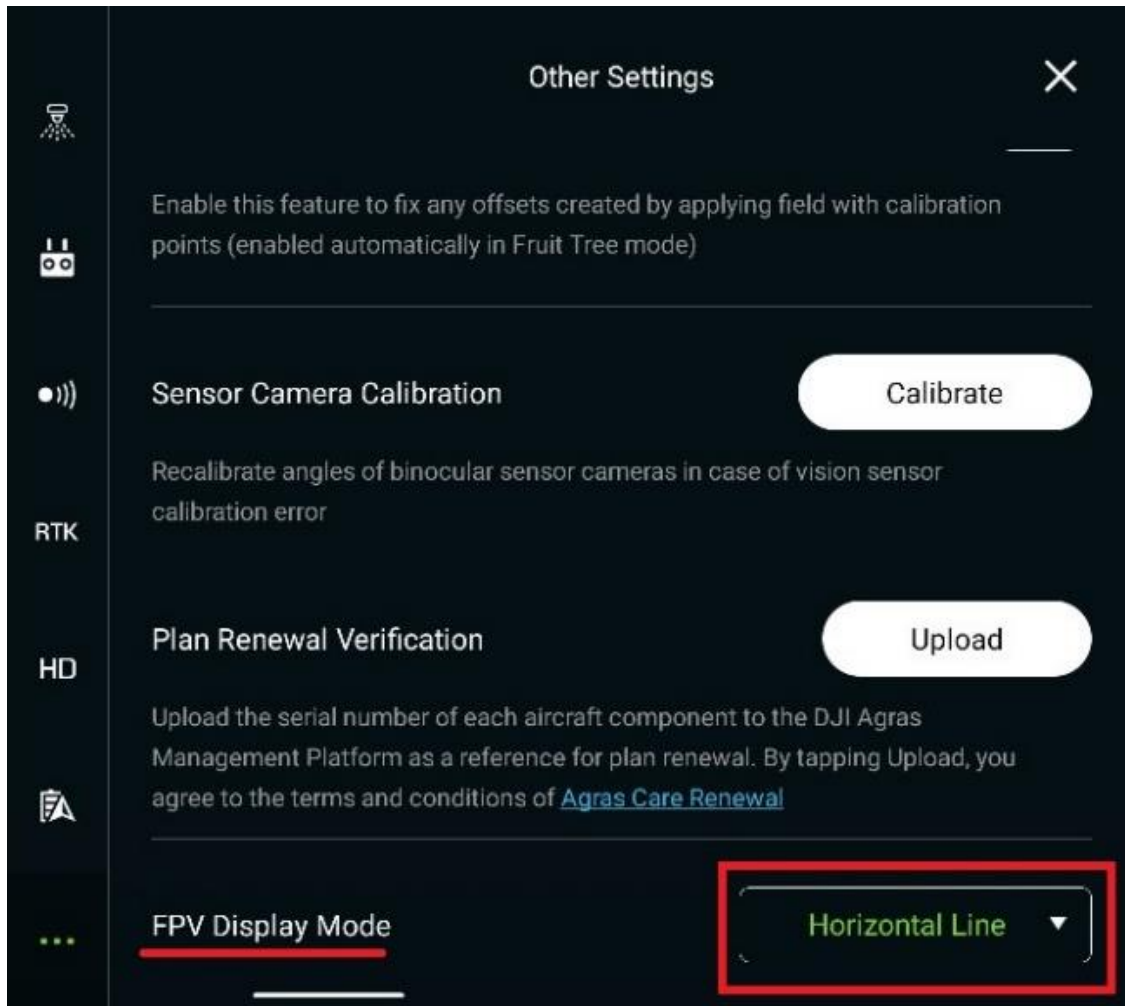


14. Vision Calibration by Using DJI Agras App

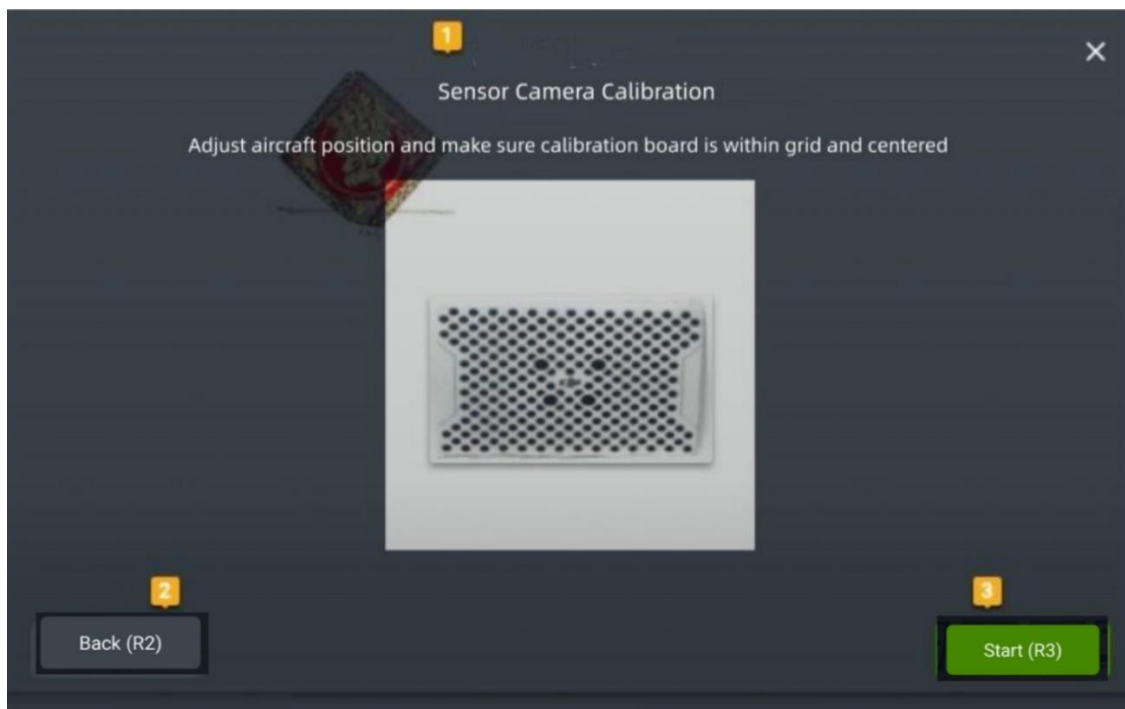
1. If a “Binocular vision sensor error. Auto Obstacle Avoidance unavailable. Contact local dealer to calibrate sensor” warning is shown in DJI Agras app, please contact your local dealer for vision calibration.
2. Go to the DJI Agras app and enter “**Other Settings**”, then tap “**Sensor Camera Calibration**”.
3. Fix the calibration board on the wall according to the position shown in the figure and app on-screen instructions, and place the aircraft on ground and in front of calibration board. Make the camera face the calibration board and keep a distance of **0.2 m to 0.4 m**.

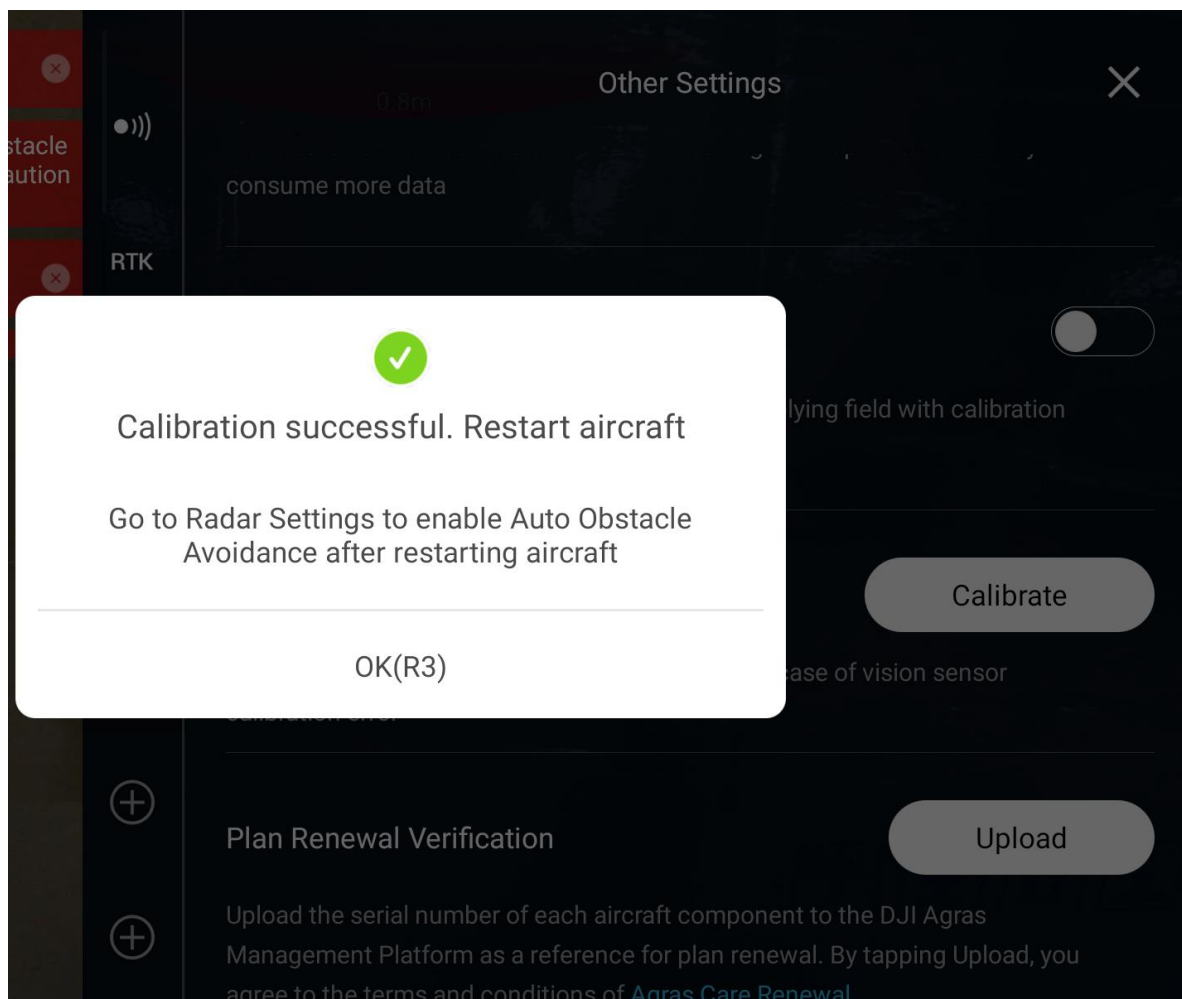


4. Set “FPV Display Mode” under Other Settings to “**Horizontal Line**”, adjust the aircraft position and make sure the calibration board is **within grid and centered**.



5. Tap “Start” and follow the on-screen instructions to complete the calibration. (Notes: The DJI logo on the calibration board must be placed forward, and cannot be placed sideways, upside down or skewed. Otherwise, the calibration fails.





22. Applying Glue to T40/T20P Aircraft Arm Locking Piece Connection Rod

- **Related Product:** T40/T20P
- **Troubleshooting:**
 1. Check whether the red screw glue is applied to the **connection rods** on the four aircraft arms. If not, disassemble the aircraft arm and apply the glue again.

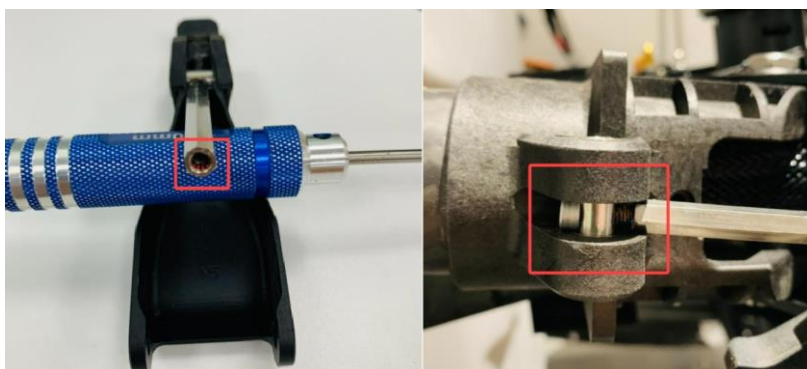


- **How to apply the glue:**

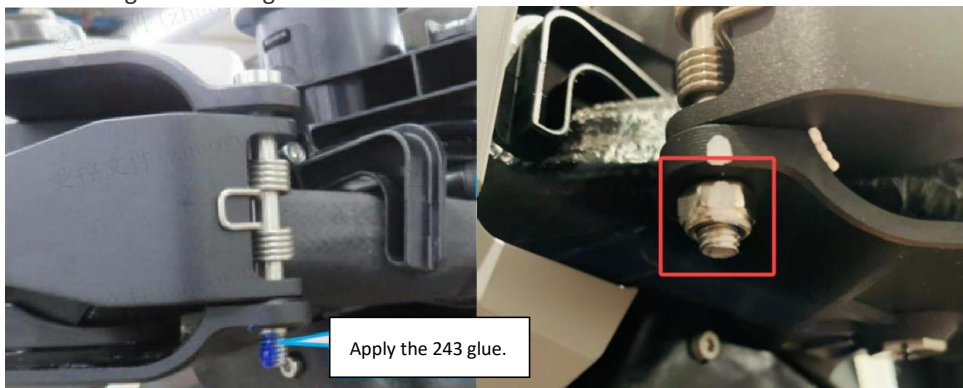
1. Apply an appropriate amount of screw glue (263 red screw glue) to the thread of the connection rod.
2. Mount the connection rod to the handle.



3. Apply an appropriate amount of screw glue (263 red screw glue) to the end part of the connection rod.
4. Align the screw with the hole on the connection rod and tighten the screw.



5. Apply an appropriate amount of screw glue (243 blue screw glue) to the bottom of the fixing bolt.
6. Attach the gasket and tighten the nut.



Apply the 243 glue.

7. 263 red screw glue and 243 blue screw glue are shown as below:



23. T20P Centrifugal Spray Rod Coming off

- **Phenomenon:** The T20P centrifugal spray rod comes off.
- **Solution:** Install the damping rubber ring (Material Number: YC.JG.MY000814.02) to the fixing nut of the spray rod, as shown below:



24. What should I do there is remaining liquid in the T40/T20P spray tank after spraying?

- **Phenomenon:** There is remaining liquid in the T40/T20P spray tank after spraying.
- **Solution:** The new operating logic is that there is remaining liquid (about 0.5L) in the T40/T20P spray tank after spraying, but the remaining liquid payload shown in the app will be 0. The remaining liquid need to be drained out manually. This is to **protect the delivery pump and dual atomized sprinkler**. If the delivery pump is running without water, there is a high probability that the pump will be damaged and worn. It is normal that there will be remaining liquid in the T20P spray tank after spraying.

25. T40/T20P “Pump %d Flow Calculation Error” App Warning or Inaccurate Spray Rate (Not Reach to the Target)

- **Phenomenon:** The Agras app prompts a “Pump %d Flow Calculation Error” warning or the spray rate is incorrect that does not reach to the target.
- Reason and Troubleshooting Plans

Reason	Troubleshooting Plans
The inner part of the impeller pump is worn or scorched, or the shaft is broken.	Disassemble the impeller pump and check whether the shaft is broken and whether the impeller is worn seriously. If yes, replace the pump.
The one-way valve cannot be opened fully.	Replace the valve.
The hose with bellows above the centrifugal sprinkler is bent.	Remove the hose and check whether it is bent. If yes, replace the hose.
The sprinkler is worn or damaged.	Disassemble the sprinkler and check whether it is worn, scorched or damaged. If yes, replace the sprinkler.

